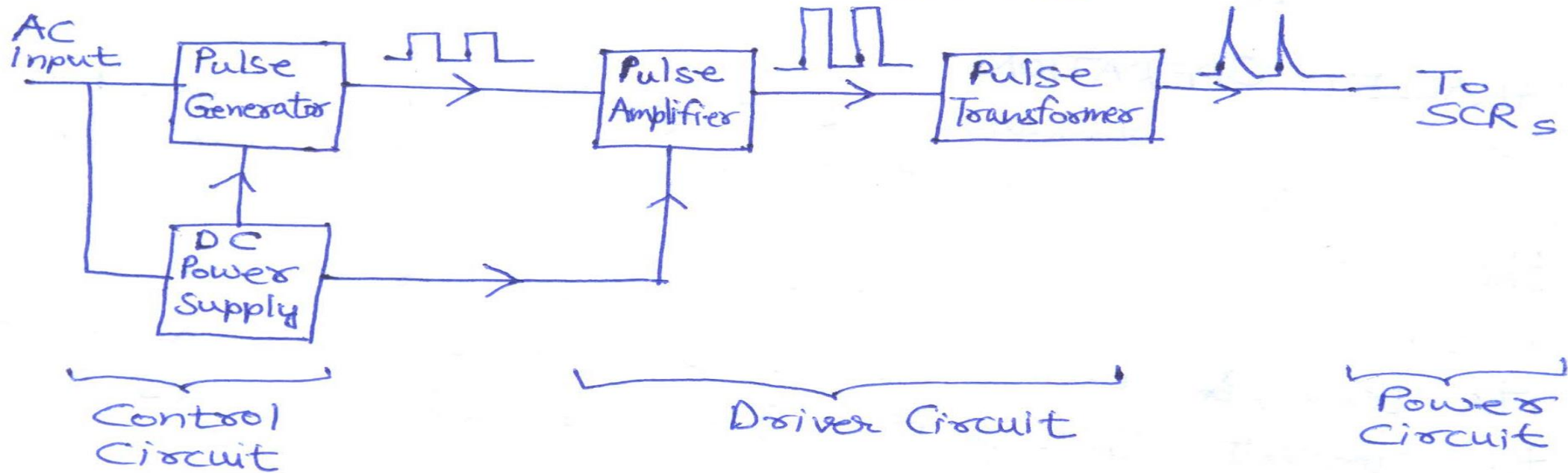


Firing Circuits of SCR and TRIAC

The most common method for controlling the onset of conduction in an SCR is by means of gate voltage control. This method can be accurately used for turning on the SCR at the desired instant of time. The gate control circuit is also called firing or triggering circuit.

MAIN FEATURES OF FIRING CIRCUITS

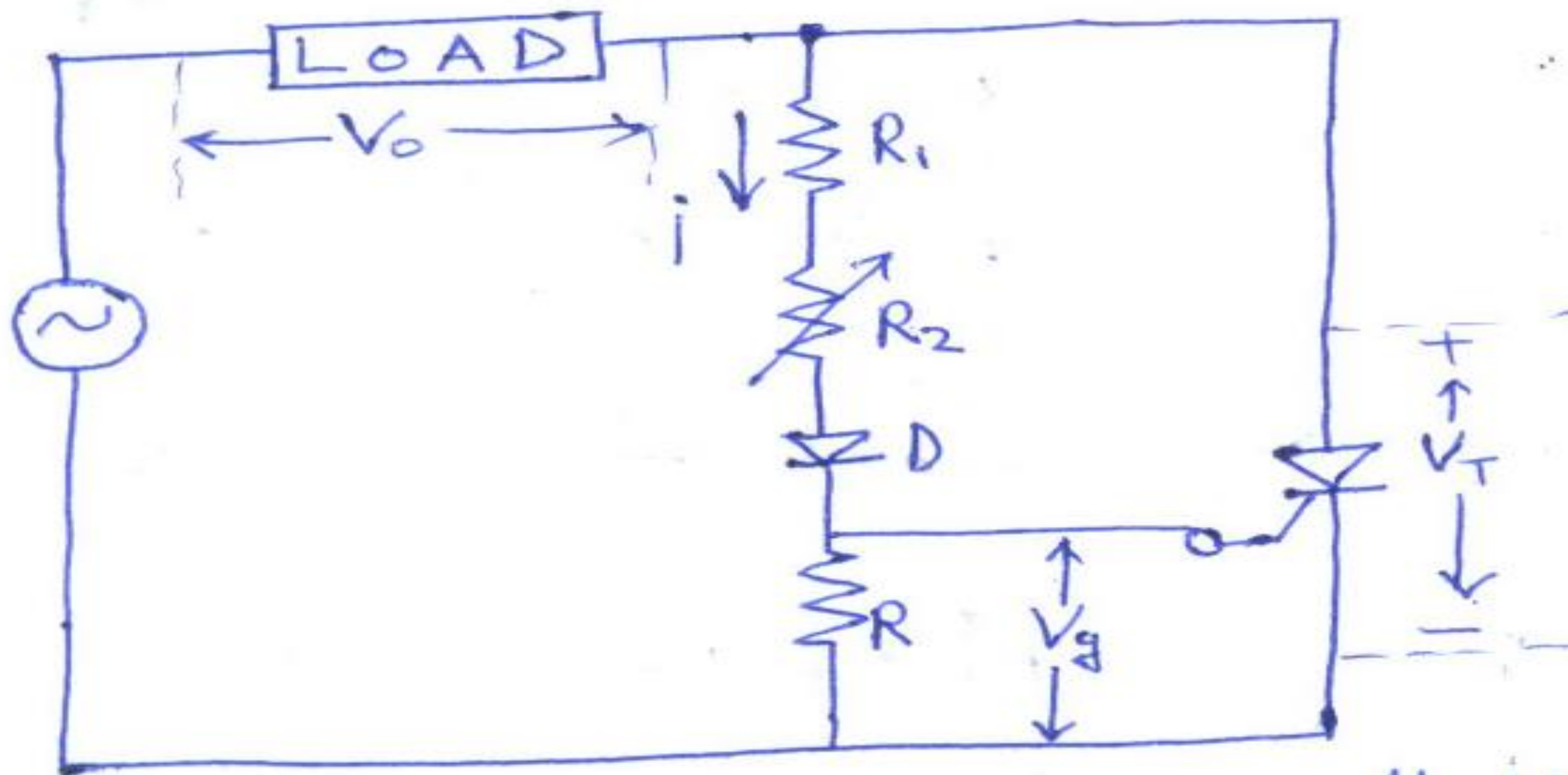


* Function of pulse transformer is isolation.

A firing circuit should fulfill the following two conditions:-

- (i) If power circuit has more than one SCR, the firing circuit should produce gating pulses for each SCR at the desired instant. These pulses must be periodic in nature & the sequence of firing must correspond with the type of thyristorised power controller.
- (ii) The control signal generated by a firing circuit should be of sufficient magnitude to turn on the SCR. For ensuring certainty of turn on, the voltage pulses should be fed to a driver circuit.

R Firing Circuits



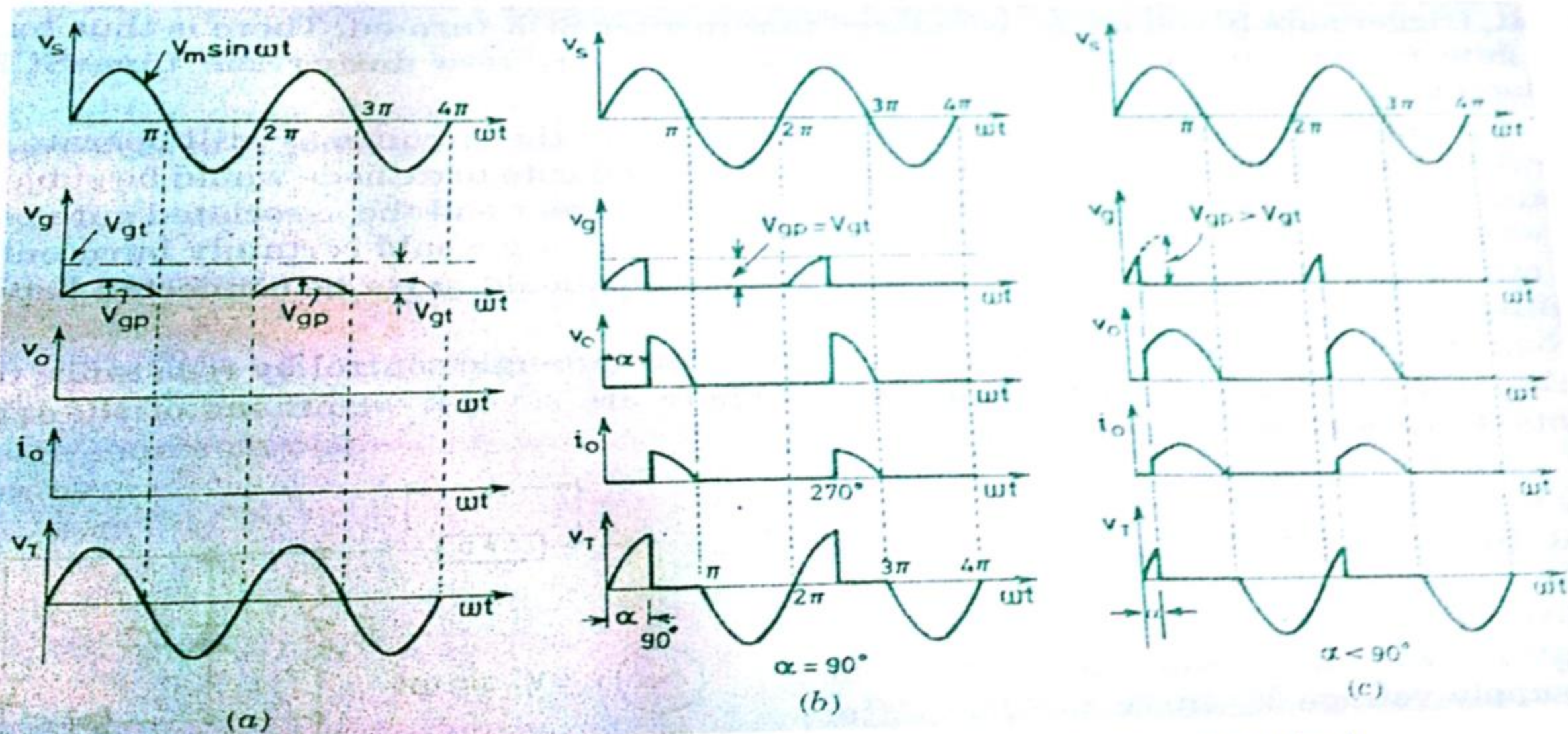


Fig. 4.65. Resistance firing of an SCR in a half-wave circuit with dc load
 (a) No triggering of SCR (b) $\alpha = 90^\circ$ (c) $\alpha < 90^\circ$.

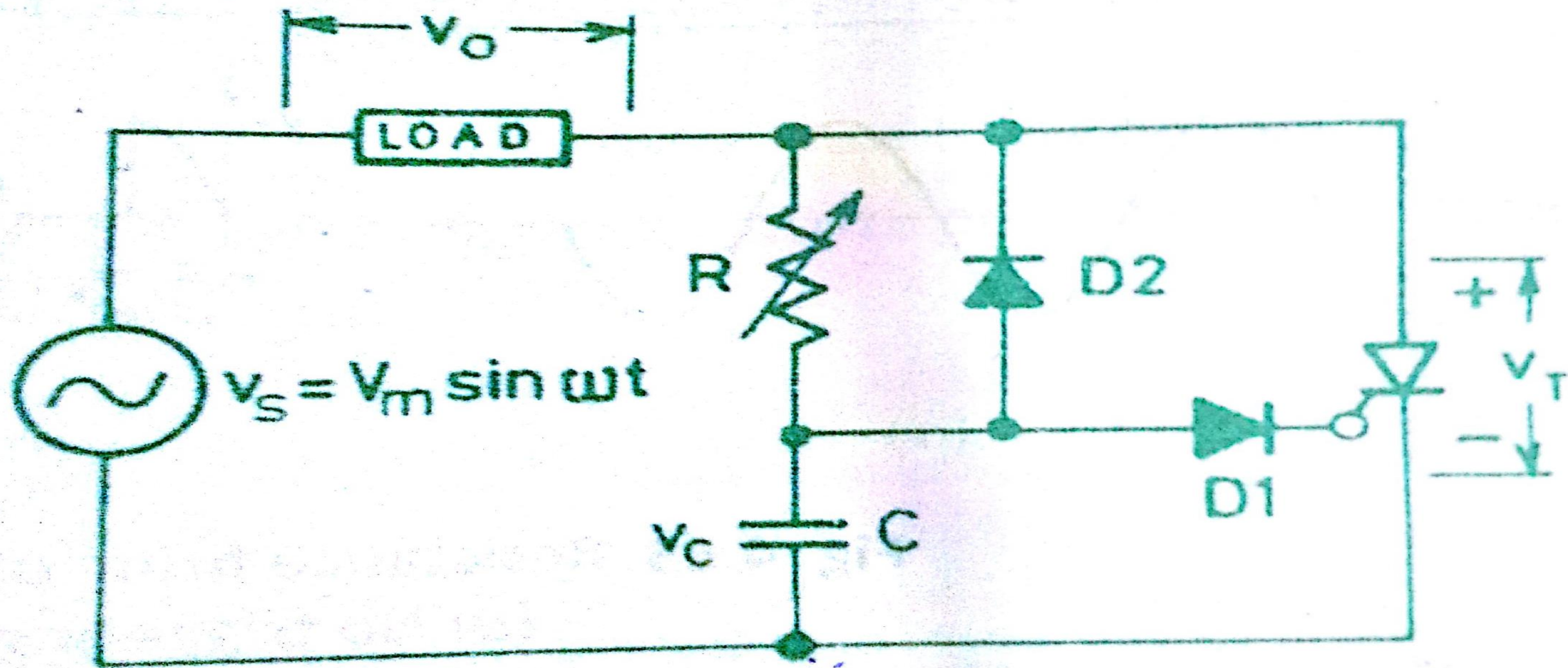
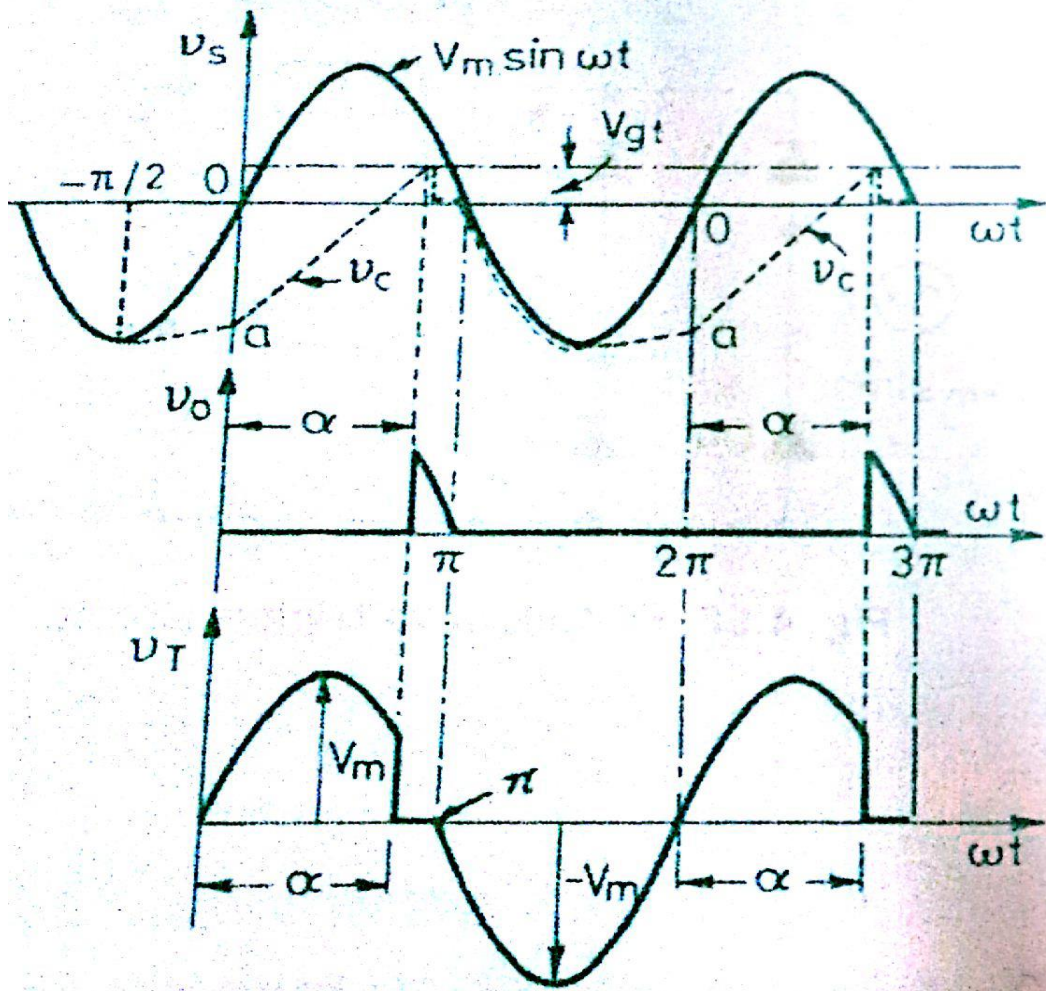
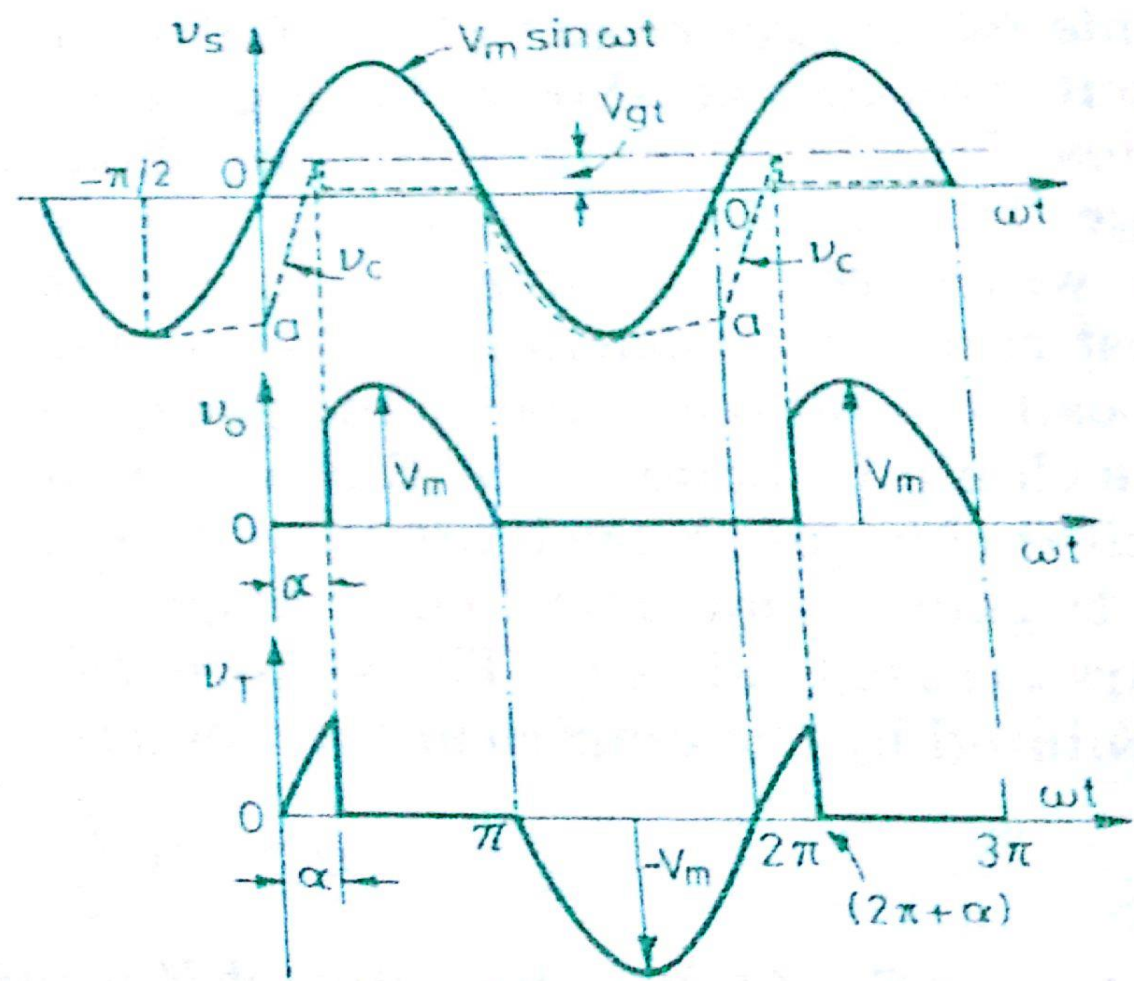


Fig. 4.66. RC half-wave trigger circuit.



(a)

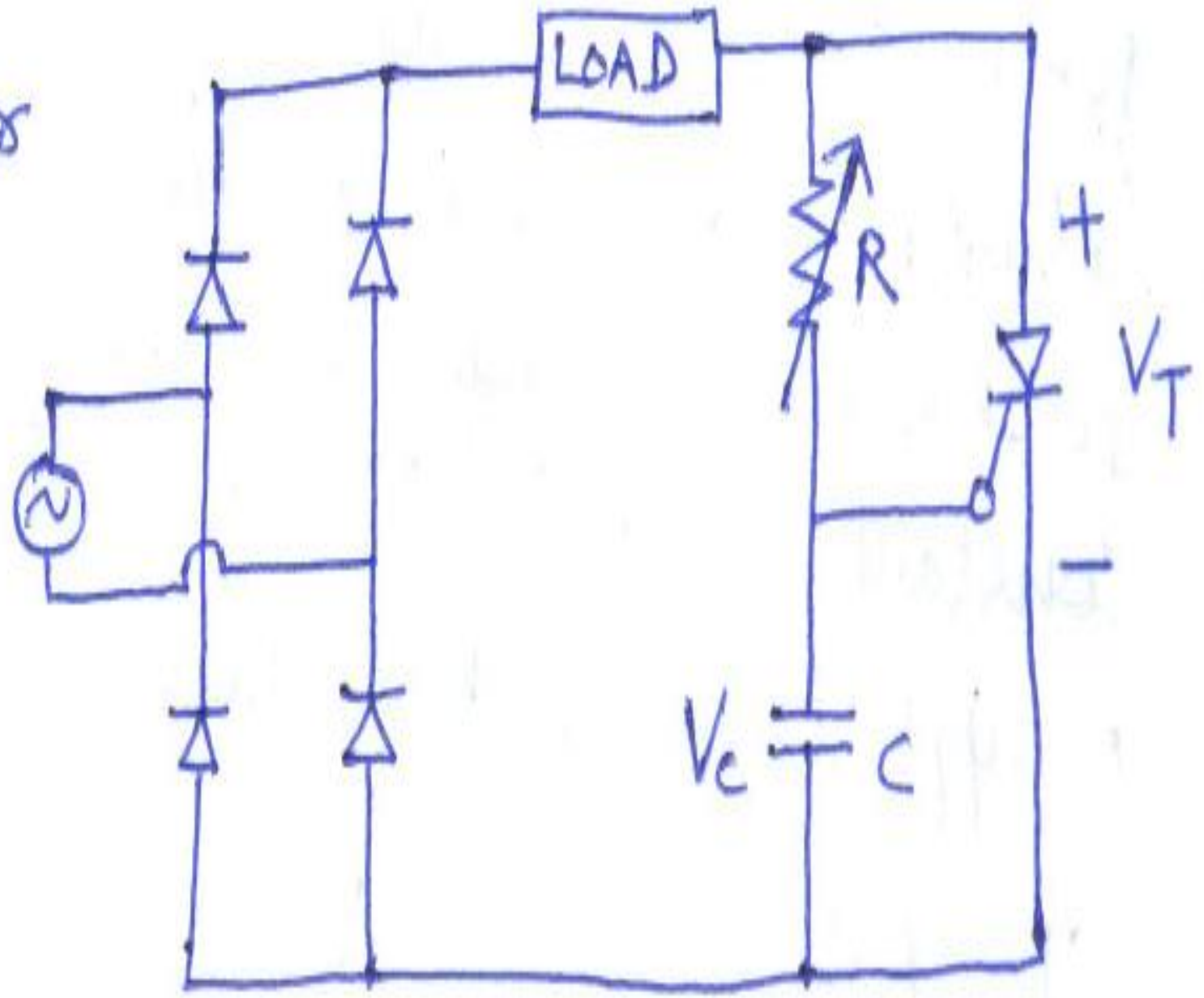


(b)

4.67. Waveforms for RC half-wave trigger circuit of Fig. 4.66 (a) high value of R (b) low value of R .

1

RC Full Wave Trigger Circuit



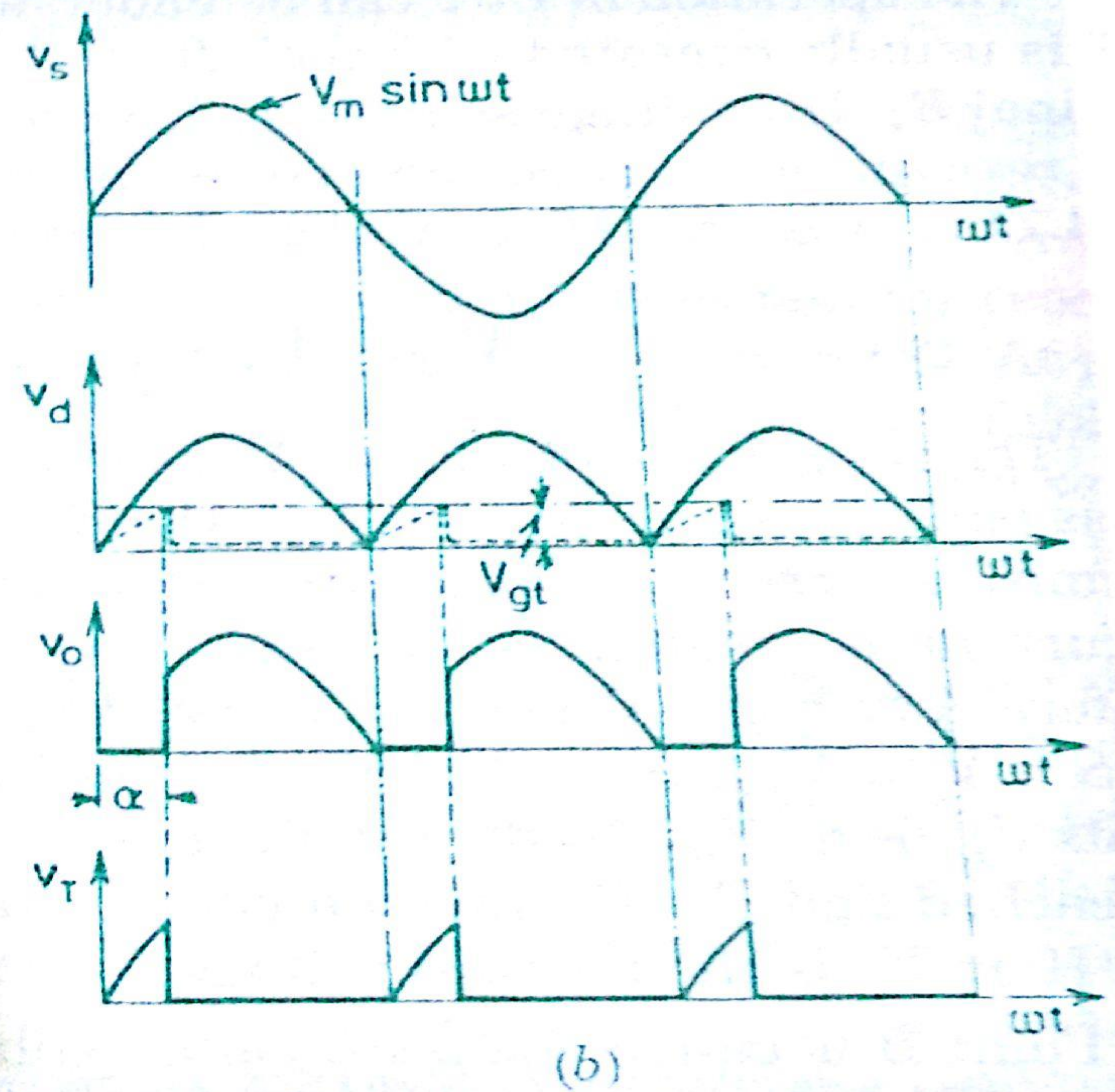
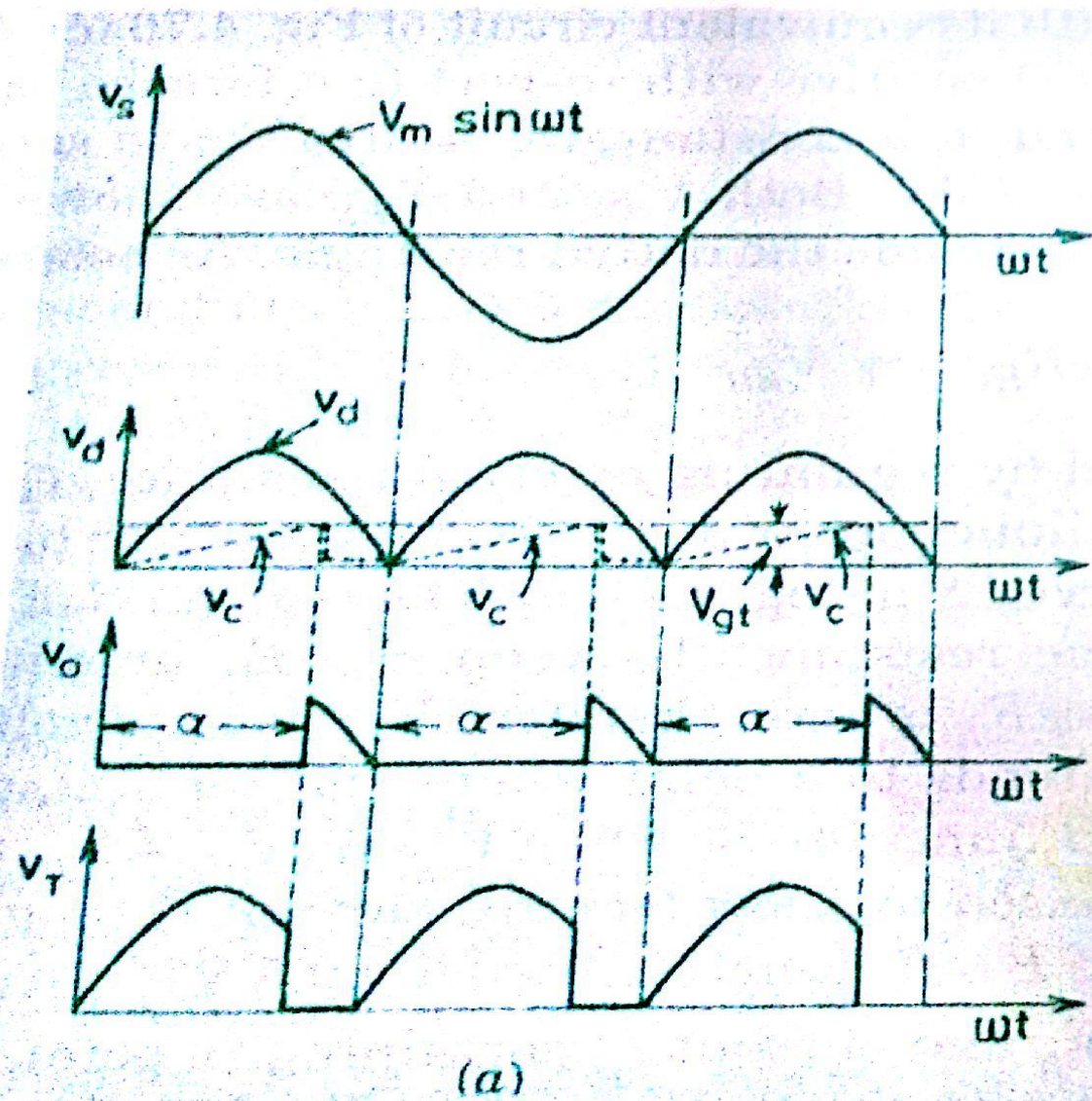


Fig. 4.69. Waveforms for RC half-wave trigger circuit of Fig. 4.68 (a) high value of R (b) low value of R .